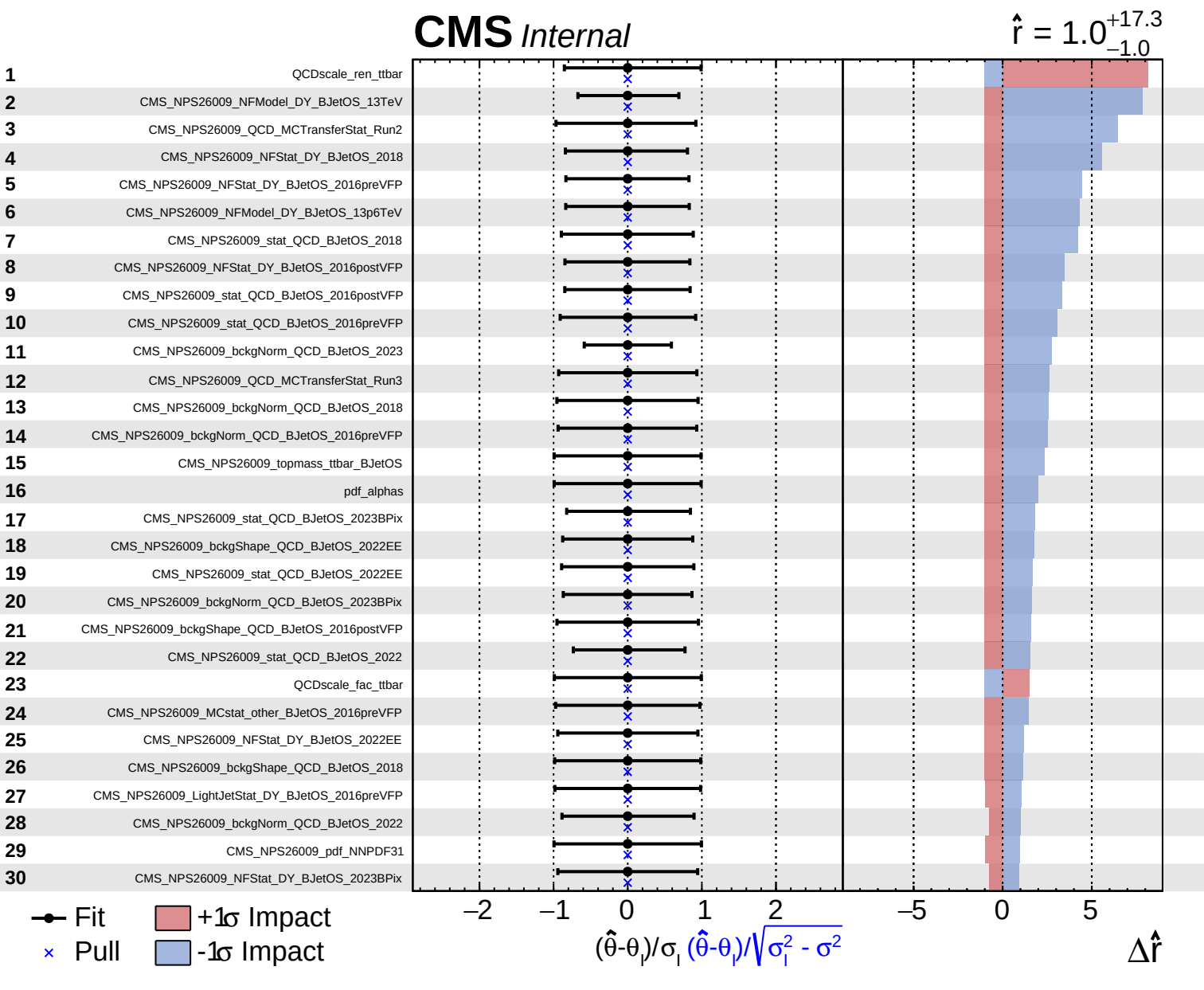
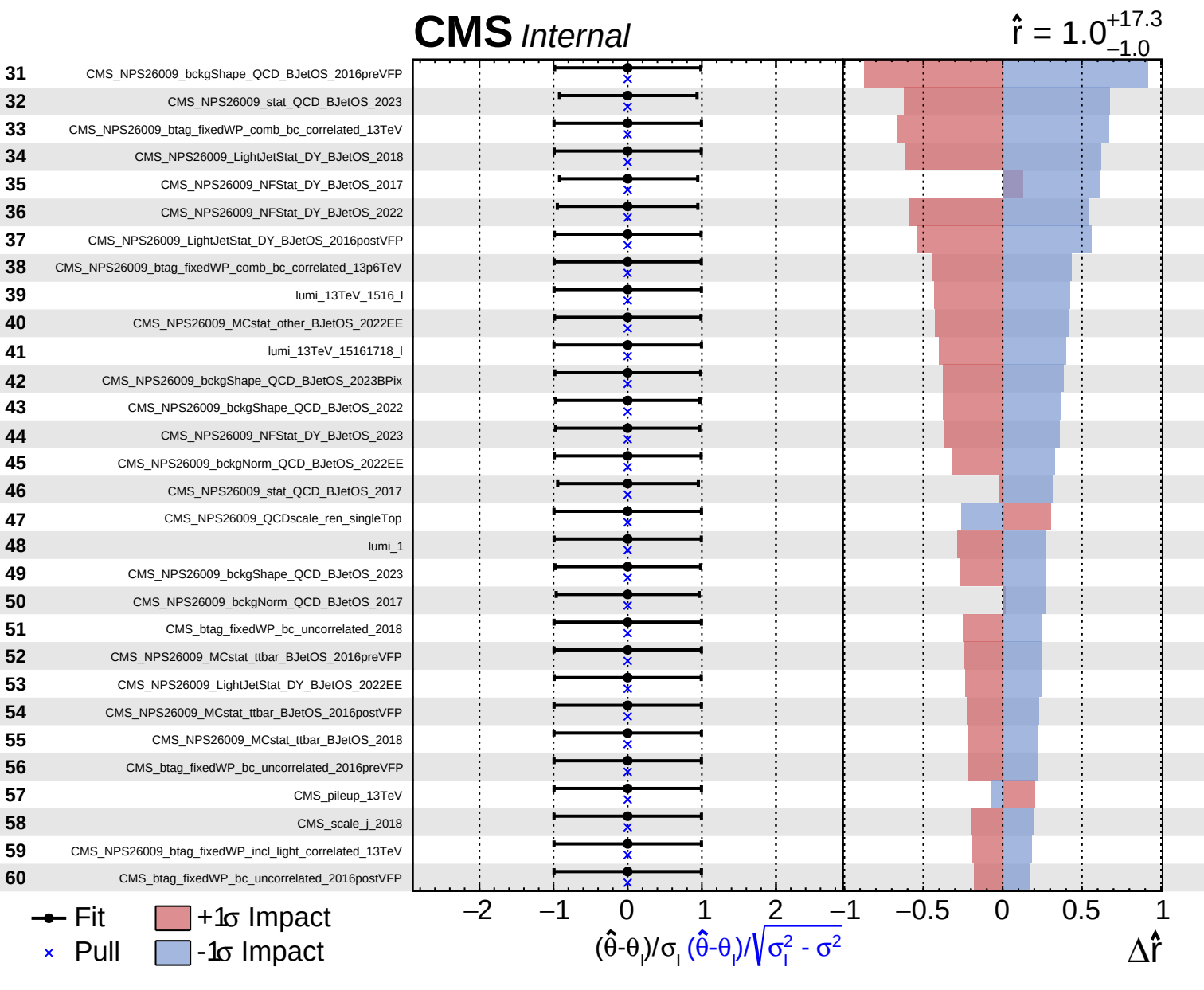
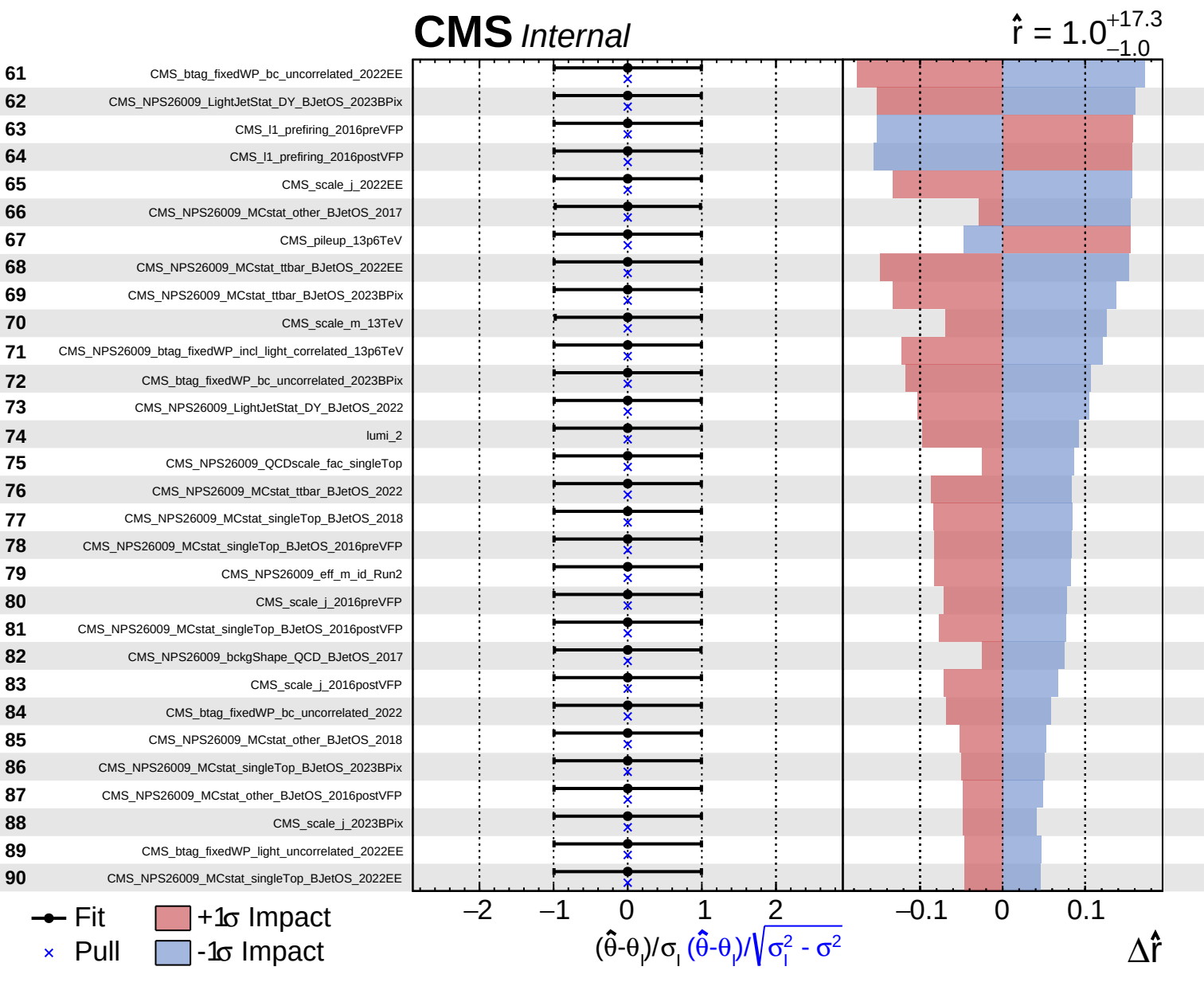
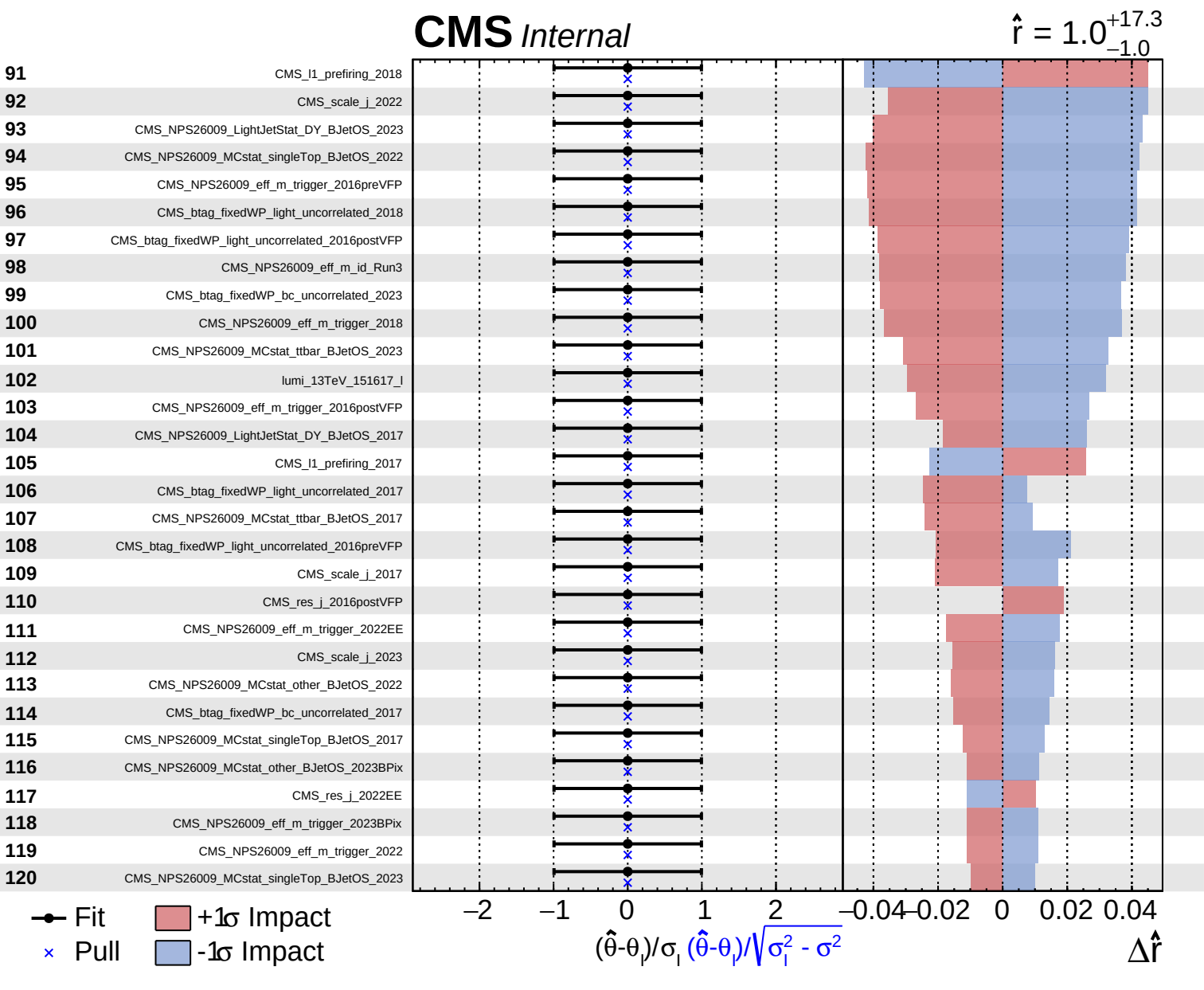
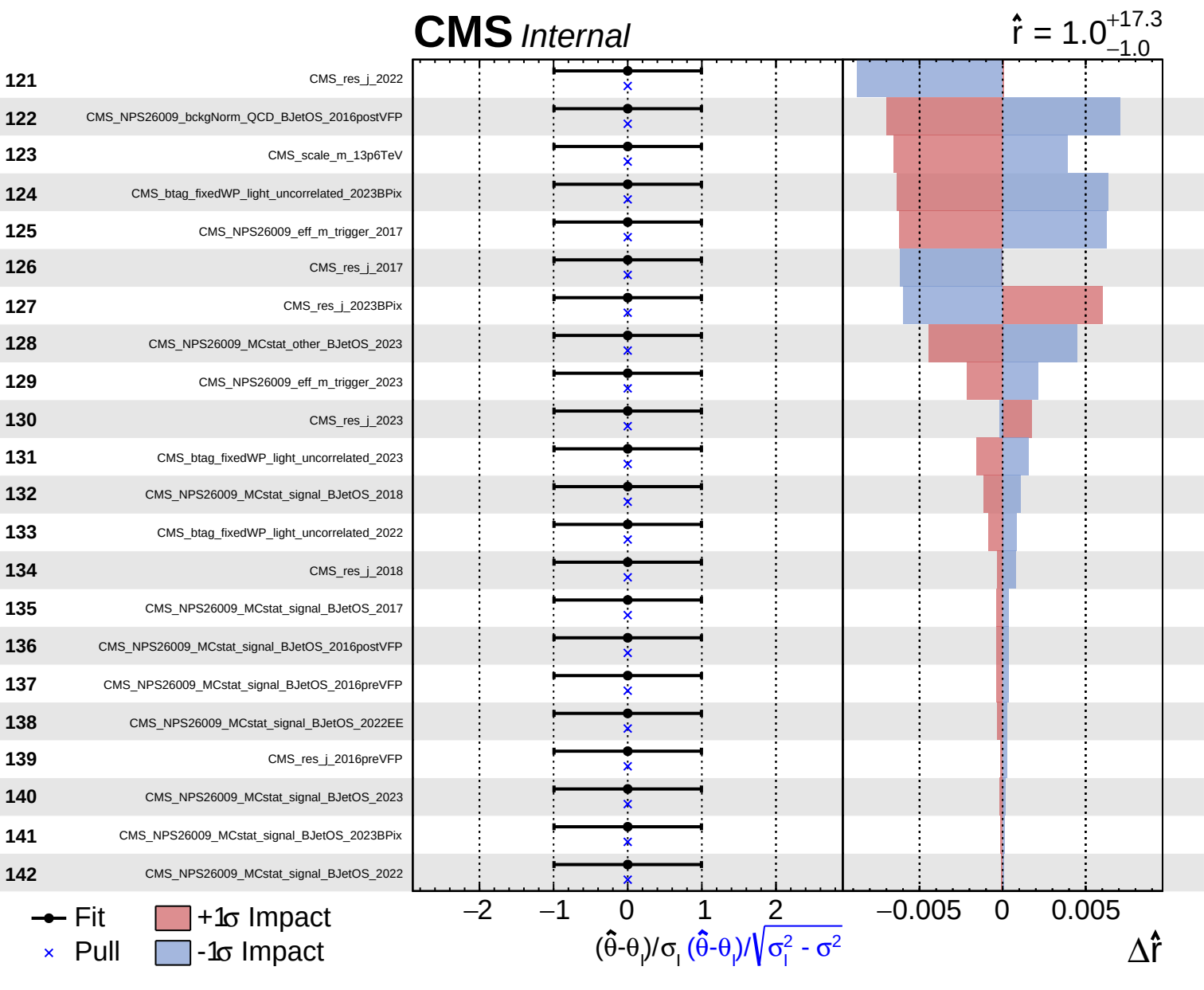

CMS Internal
$\hat{r} = 1.0^{+17.3}_{-1.0}$
1 QCDscale_ren_ttbar
2 CMS_NPS26009_NFModel_DY_BJetOS_13TeV
3 CMS_NPS26009_QCD_MCTransferStat_Run2
4 CMS_NPS26009_NFStat_DY_BJetOS_2018
5 CMS_NPS26009_NFStat_DY_BJetOS_2016preVFP
6 CMS_NPS26009_NFModel_DY_BJetOS_13p6TeV
7 CMS_NPS26009_stat_QCD_BJetOS_2018
8 CMS_NPS26009_NFStat_DY_BJetOS_2016postVFP
9 CMS_NPS26009_stat_QCD_BJetOS_2016postVFP
10 CMS_NPS26009_stat_QCD_BJetOS_2016preVFP
11 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023
12 CMS_NPS26009_QCD_MCTransferStat_Run3
13 CMS_NPS26009_bckgNorm_QCD_BJetOS_2018
14 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016preVFP
15 CMS_NPS26009_topmass_ttbar_BJetOS
16 pdf_alphas
17 CMS_NPS26009_stat_QCD_BJetOS_2023BPix
18 CMS_NPS26009_bckgShape_QCD_BJetOS_2022EE
19 CMS_NPS26009_stat_QCD_BJetOS_2022EE
20 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023BPix
21 CMS_NPS26009_bckgShape_QCD_BJetOS_2016postVFP
22 CMS_NPS26009_stat_QCD_BJetOS_2022
23 QCDscale_fac_ttbar
24 CMS_NPS26009_MCstat_other_BJetOS_2016preVFP
25 CMS_NPS26009_NFStat_DY_BJetOS_2022EE
26 CMS_NPS26009_bckgShape_QCD_BJetOS_2018
27 CMS_NPS26009_LightJetStat_DY_BJetOS_2016preVFP
28 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022
29 CMS_NPS26009_pdf_NNPDF31
30 CMS_NPS26009_NFStat_DY_BJetOS_2023BPix
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
$(\hat{\theta}-\theta_I)/\sigma_I$
−5 0 5
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 1.0^{+17.3}_{-1.0}$
31 CMS_NPS26009_bckgShape_QCD_BJetOS_2016preVFP
32 CMS_NPS26009_stat_QCD_BJetOS_2023
33 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13TeV
34 CMS_NPS26009_LightJetStat_DY_BJetOS_2018
35 CMS_NPS26009_NFStat_DY_BJetOS_2017
36 CMS_NPS26009_NFStat_DY_BJetOS_2022
37 CMS_NPS26009_LightJetStat_DY_BJetOS_2016postVFP
38 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13p6TeV
39 lumi_13TeV_1516_l
40 CMS_NPS26009_MCstat_other_BJetOS_2022EE
41 lumi_13TeV_15161718_l
42 CMS_NPS26009_bckgShape_QCD_BJetOS_2023BPix
43 CMS_NPS26009_bckgShape_QCD_BJetOS_2022
44 CMS_NPS26009_NFStat_DY_BJetOS_2023
45 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022EE
46 CMS_NPS26009_stat_QCD_BJetOS_2017
47 CMS_NPS26009_QCDscale_ren_singleTop
48 lumi_1
49 CMS_NPS26009_bckgShape_QCD_BJetOS_2023
50 CMS_NPS26009_bckgNorm_QCD_BJetOS_2017
51 CMS_btag_fixedWP_bc_uncorrelated_2018
52 CMS_NPS26009_MCstat_ttbar_BJetOS_2016preVFP
53 CMS_NPS26009_LightJetStat_DY_BJetOS_2022EE
54 CMS_NPS26009_MCstat_ttbar_BJetOS_2016postVFP
55 CMS_NPS26009_MCstat_ttbar_BJetOS_2018
56 CMS_btag_fixedWP_bc_uncorrelated_2016preVFP
57 CMS_pileup_13TeV
58 CMS_scale_j_2018
59 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13TeV
60 CMS_btag_fixedWP_bc_uncorrelated_2016postVFP
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
$(\hat{\theta}-\theta_I)/\sigma_I$
−1 −0.5 0 0.5 1
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 1.0^{+17.3}_{-1.0}$
61 CMS_btag_fixedWP_bc_uncorrelated_2022EE
62 CMS_NPS26009_LightJetStat_DY_BJetOS_2023BPix
63 CMS_l1_prefiring_2016preVFP
64 CMS_l1_prefiring_2016postVFP
65 CMS_scale_j_2022EE
66 CMS_NPS26009_MCstat_other_BJetOS_2017
67 CMS_pileup_13p6TeV
68 CMS_NPS26009_MCstat_ttbar_BJetOS_2022EE
69 CMS_NPS26009_MCstat_ttbar_BJetOS_2023BPix
70 CMS_scale_m_13TeV
71 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13p6TeV
72 CMS_btag_fixedWP_bc_uncorrelated_2023BPix
73 CMS_NPS26009_LightJetStat_DY_BJetOS_2022
74 lumi_2
75 CMS_NPS26009_QCDscale_fac_singleTop
76 CMS_NPS26009_MCstat_ttbar_BJetOS_2022
77 CMS_NPS26009_MCstat_singleTop_BJetOS_2018
78 CMS_NPS26009_MCstat_singleTop_BJetOS_2016preVFP
79 CMS_NPS26009_eff_m_id_Run2
80 CMS_scale_j_2016preVFP
81 CMS_NPS26009_MCstat_singleTop_BJetOS_2016postVFP
82 CMS_NPS26009_bckgShape_QCD_BJetOS_2017
83 CMS_scale_j_2016postVFP
84 CMS_btag_fixedWP_bc_uncorrelated_2022
85 CMS_NPS26009_MCstat_other_BJetOS_2018
86 CMS_NPS26009_MCstat_singleTop_BJetOS_2023BPix
87 CMS_NPS26009_MCstat_other_BJetOS_2016postVFP
88 CMS_scale_j_2023BPix
89 CMS_btag_fixedWP_light_uncorrelated_2022EE
90 CMS_NPS26009_MCstat_singleTop_BJetOS_2022EE
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
−0.1
0
0.1
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 1.0^{+17.3}_{-1.0}$
91 CMS_l1_prefiring_2018
92 CMS_scale_j_2022
93 CMS_NPS26009_LightJetStat_DY_BJetOS_2023
94 CMS_NPS26009_MCstat_singleTop_BJetOS_2022
95 CMS_NPS26009_eff_m_trigger_2016preVFP
96 CMS_btag_fixedWP_light_uncorrelated_2018
97 CMS_btag_fixedWP_light_uncorrelated_2016postVFP
98 CMS_NPS26009_eff_m_id_Run3
99 CMS_btag_fixedWP_bc_uncorrelated_2023
100 CMS_NPS26009_eff_m_trigger_2018
101 CMS_NPS26009_MCstat_ttbar_BJetOS_2023
102 lumi_13TeV_151617_l
103 CMS_NPS26009_eff_m_trigger_2016postVFP
104 CMS_NPS26009_LightJetStat_DY_BJetOS_2017
105 CMS_l1_prefiring_2017
106 CMS_btag_fixedWP_light_uncorrelated_2017
107 CMS_NPS26009_MCstat_ttbar_BJetOS_2017
108 CMS_btag_fixedWP_light_uncorrelated_2016preVFP
109 CMS_scale_j_2017
110 CMS_res_j_2016postVFP
111 CMS_NPS26009_eff_m_trigger_2022EE
112 CMS_scale_j_2023
113 CMS_NPS26009_MCstat_other_BJetOS_2022
114 CMS_btag_fixedWP_bc_uncorrelated_2017
115 CMS_NPS26009_MCstat_singleTop_BJetOS_2017
116 CMS_NPS26009_MCstat_other_BJetOS_2023BPix
117 CMS_res_j_2022EE
118 CMS_NPS26009_eff_m_trigger_2023BPix
119 CMS_NPS26009_eff_m_trigger_2022
120 CMS_NPS26009_MCstat_singleTop_BJetOS_2023
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.04 −0.02 0 0.02 0.04
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 1.0^{+17.3}_{-1.0}$
121 CMS_res_j_2022
122 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016postVFP
123 CMS_scale_m_13p6TeV
124 CMS_btag_fixedWP_light_uncorrelated_2023BPix
125 CMS_NPS26009_eff_m_trigger_2017
126 CMS_res_j_2017
127 CMS_res_j_2023BPix
128 CMS_NPS26009_MCstat_other_BJetOS_2023
129 CMS_NPS26009_eff_m_trigger_2023
130 CMS_res_j_2023
131 CMS_btag_fixedWP_light_uncorrelated_2023
132 CMS_NPS26009_MCstat_signal_BJetOS_2018
133 CMS_btag_fixedWP_light_uncorrelated_2022
134 CMS_res_j_2018
135 CMS_NPS26009_MCstat_signal_BJetOS_2017
136 CMS_NPS26009_MCstat_signal_BJetOS_2016postVFP
137 CMS_NPS26009_MCstat_signal_BJetOS_2016preVFP
138 CMS_NPS26009_MCstat_signal_BJetOS_2022EE
139 CMS_res_j_2016preVFP
140 CMS_NPS26009_MCstat_signal_BJetOS_2023
141 CMS_NPS26009_MCstat_signal_BJetOS_2023BPix
142 CMS_NPS26009_MCstat_signal_BJetOS_2022
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.005
0
0.005
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$